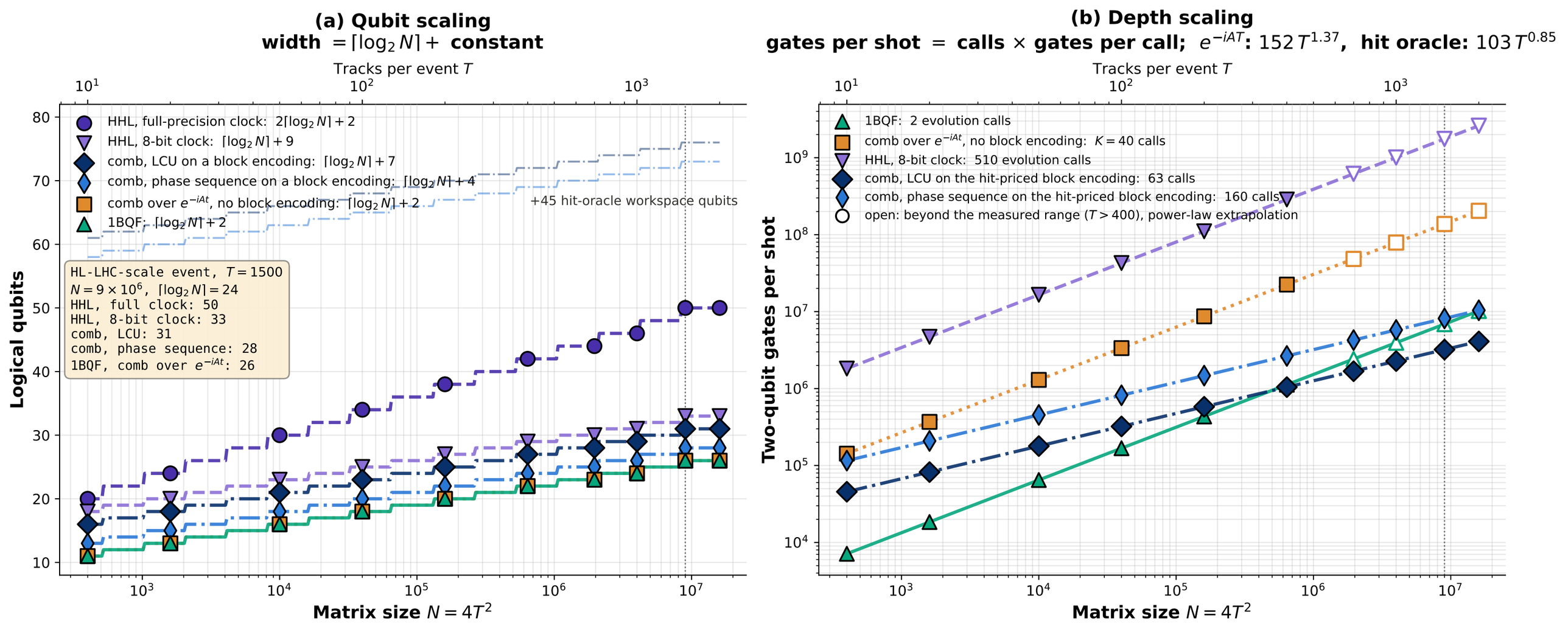

(a) Qubit scaling
width = ⌈log2 N⌉ + constant
Tracks per event T
Logical qubits
Matrix size N = 4T^2
HHL, full-precision clock: 2⌈log2 N⌉ + 2
HHL, 8-bit clock: ⌈log2 N⌉ + 9
comb, LCU on a block encoding: ⌈log2 N⌉ + 7
comb, phase sequence on a block encoding: ⌈log2 N⌉ + 4
comb over e^{-iAt}, no block encoding: ⌈log2 N⌉ + 2
1BQF: ⌈log2 N⌉ + 2
+45 hit-oracle workspace qubits
HL-LHC-scale event, T = 1500
N = 9 × 10^6, ⌈log2 N⌉ = 24
HHL, full clock: 50
HHL, 8-bit clock: 33
comb, LCU: 31
comb, phase sequence: 28
1BQF, comb over e^{-iAt}: 26
(b) Depth scaling
gates per shot = calls × gates per call; e^{-iAT}: 152 T^1.37, hit oracle: 103 T^0.85
Tracks per event T
Two-qubit gates per shot
Matrix size N = 4T^2
1BQF: 2 evolution calls
comb over e^{-iAt}, no block encoding: K = 40 calls
HHL, 8-bit clock: 510 evolution calls
comb, LCU on the hit-priced block encoding: 63 calls
comb, phase sequence on the hit-priced block encoding: 160 calls
open: beyond the measured range (T > 400), power-law extrapolation